Northern University of Business and Technology, Khulna
Department of Computer Science and Engineering
CSE 4104 — Software Development III

WEEK 01 — Team Formation & Initial Project Idea Selection

Team: CSE4104-7D-T06 | Section: 7D | 2 November 2026
Supervised By: Md. Riaz Mahmud, Assistant Professor, CSE Dept., NUBTK

1. Official Team Information

Official Team Name	CSE4104-7D-T06
Section	7D
Course Code	CSE 4104
Course Title	Software Development III
Project Title (Initial Idea)	AI Academic Mentor-Personalized Learning & Study Assistant
Supervised By	Md. Riaz Mahmud, Assistant Professor, CSE Dept., NUBTK
Submission Week	Week 1 — Team Formation & Idea Selection
Submission Date	2 November 2026

2. Team Leader Information

Team Leader Name	Efaz Rahman
Team Leader Student ID	11230121162
Role	Team Leader — Coordination, task management, weekly reports, GitHub management, communication with instructor

3. Full Member List with IDs

#	Name	Student ID	Role	Responsibility
1	Efaz Rahman	11230121162	Team Leader	Coordination, task management, GitHub management
2	Santu Mondal	11230321312	Frontend Developer	UI design, React components, responsive design
3	MD Ruhan Tanveer	11230321335	Backend Developer	REST APIs, authentication, database design

4. Project Title (Initial Idea)

AI Academic Mentor— Personalized Learning & Study Assistant

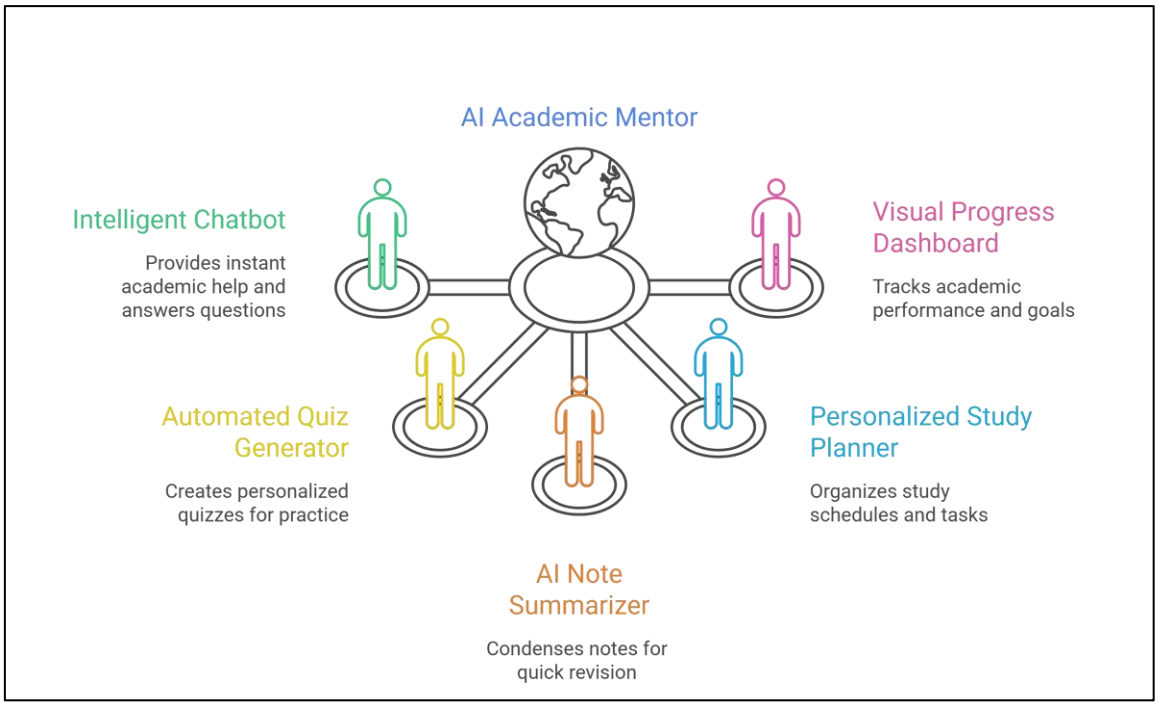
5. Preliminary Project Description

AI Academic Mentor is a full-stack, AI-integrated web application designed to function as a personalized study assistant for university students. The platform addresses real academic challenges that students in Bangladesh face daily — unorganized study schedules, lack of personalized learning support, no instant academic help outside class hours, and time wasted searching for resources.

The system will combine three core AI features — an intelligent chatbot, an automated quiz generator, and an AI note summarizer — alongside a personalized study planner and visual progress dashboard into a single cohesive platform.

Target Users:

- University students who need organized, intelligent, and personalized study support.
- Students preparing for exams who need quick quiz practice and note revision tools.
- Students who lack access to tutors and need 24/7 AI-powered academic assistance.



6. Why AI Is Needed in This Project

Current learning management systems and note-taking apps are entirely static — they store and display content but do not adapt to the individual student. AI is the essential layer that enables:

- Personalized recommendations based on each student's performance and identified weak areas.
- An AI chatbot that provides instant, contextual, and accurate academic answers 24/7 without requiring instructor availability.
- Automated quiz generation from any topic, saving significant time that students would otherwise spend creating practice materials manually.
- AI-powered summarization of dense study notes into concise, easy-to-review content.
- Intelligent study plan generation based on a student's subjects, goals, and available study hours.

Without AI, this system would be just another CRUD dashboard. The AI integration is what transforms it into a genuinely useful, adaptive academic tool.

7. Initial Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	User interface and responsive design
Backend	Node.js + Express.js	REST API, business logic, authentication
Database	MongoDB + MongoDB Atlas	Flexible data storage and cloud hosting
AI Tools	Gemini API / OpenAI API	Chatbot, quiz generation, note summarization
Authentication	JWT + bcrypt	Secure login and session management
Frontend Deploy	Vercel	Fast, free frontend hosting
Backend Deploy	Render	Free backend API hosting
Version Control	GitHub	Code management and collaboration

8. Research on Similar Existing Systems

Before finalizing the project idea, the team researched similar systems from the following sources: Google, GitHub, YouTube, research papers, and product websites. The findings are summarized below:

System	Existing Features	Limitations	AI Usage
Google Classroom	Assignment submission, resource sharing, announcements	No AI personalization, fully static, no quiz generation	None
Quizlet	Flashcard sets, basic quiz practice, study mode	No AI chatbot, no study planner, no note summarizer	Minimal (spell assist)
ChatGPT (standalone)	AI question answering, text generation	No student dashboard, no quiz tracking, no planner	Full (but not tailored for students)
Notion AI	Note-taking with AI writing assist	No quiz generation, no study planning, no academic chatbot	Partial (writing assist only)
Khan Academy Khanmigo	AI tutoring chatbot for specific subjects	Limited to Khan Academy content, no custom quiz or note summary	AI chatbot only

Key Finding: No existing system combines all three AI features (chatbot + quiz generator + note summarizer) in a single platform with a personalized student dashboard. AI Academic Mentor directly addresses this gap.

9. Project Idea Validation

What real-world problem are we solving?

University students lack an intelligent, personalized, and integrated study assistant. Existing tools are fragmented, static, and do not adapt to individual academic needs.

Who are the target users?

University students — specifically students in Bangladesh who need academic support tools that are accessible online, easy to use, and intelligent enough to adapt to their individual learning patterns.

Why is AI useful in this system?

AI enables personalization, automated quiz generation, intelligent summarization, and 24/7 chatbot support — capabilities that are fundamentally impossible without AI. Without AI this would be a static note-taking app with no real value.

How will the system improve existing solutions?

AI Academic Mentor improves on existing tools by combining chatbot, quiz generator, and summarizer in one platform, connecting all features through a unified student profile, tracking progress over time, and generating personalized study plans — none of which existing tools do together.

